



PrEP and Syphilis Model



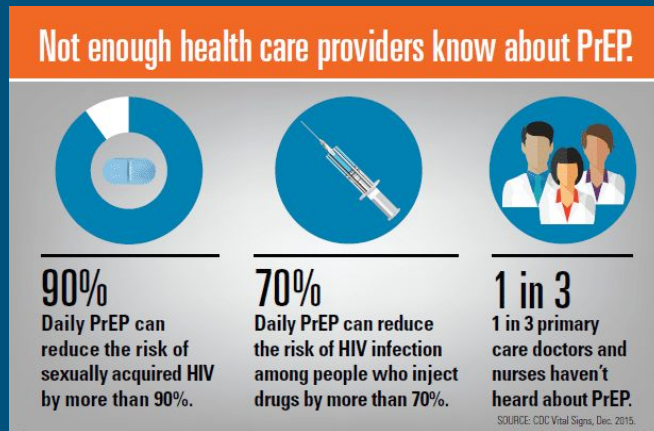
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Does the lower condom
usage by PrEP users
have an impact on the
number of syphilis
cases?

Background

- Pre-exposure Prophylaxis
- Taken by HIV-negative people
- According to the CDC:
 - 1.2 million can benefit from PrEP
 - ~360,000 use PrEP



Assumptions

- No one is partnered
- Everyone is sexually active once per week
- Sexually active population of the USA only
- Once you recover from syphilis you re-enter the susceptible population
- If you are infected from syphilis you will always use protection afterwards
- Syphilis is the only factor that will change behavior
- Constant infection/recovery times as well as transmission rates
- Condoms are the only form of protection from syphilis

PrEP vs. PEP		
PrEP and PEP are methods for preventing HIV that involve taking HIV medicines. When you take steps to protect yourself against a disease, like HIV, it's called prophylaxis. PrEP and PEP are for people who don't have HIV, but are at risk of getting it.		
PrEP stands for pre-exposure prophylaxis.	What's it called?	PEP stands for post-exposure prophylaxis.
Before HIV exposure. PrEP is taken every day, before possible exposure.	When is it taken?	After HIV exposure. In emergency situations, PEP is taken within 72 hours (3 days) after possible exposure.
PrEP is for people who don't have HIV and: • are at risk of getting HIV from sex • are at risk of getting HIV from injection drug use	Who's it for?	PEP is for people who don't have HIV but may have been exposed: • during sex • by sharing injection drug equipment • during a sexual assault • at work through a needlestick or other injury
Consistent use of PrEP can reduce the risk of getting HIV from sex by about 99% and from injection drug use by at least 74%.	How effective is it?	PEP can prevent HIV when taken correctly, but it is not always effective. Start PEP as soon as possible to give it the best chance of working.
Ask your health care provider about a prescription for PrEP , or use PreELocator.org to find a health care provider in your area who can prescribe PrEP.	How do you get it?	Within 72 hours of a potential exposure to HIV, talk to your health care provider or an emergency room doctor about a prescription for PEP .
For more information, visit 		

Variables

SPN = people susceptible to syphilis that use PrEP but no protection

SNN = people susceptible to syphilis that don't use PrEP or protection

SPP = people susceptible to syphilis that use PrEP and protection

SNP = people susceptible to syphilis that don't use PrEP but use protection

S = total susceptible population

I = people infected with syphilis

R = people recovering from syphilis

t = time in days

Parameters

$$a = \frac{mr_{none}}{Y}$$

where m = sexual encounters per day, r_{none} = the transmission probability of syphilis during unprotected sex, and Y = the total population. We made the assumption that a person would have one sexual encounter per week. Based on our research, we estimated that the transmission probability of syphilis during unprotected sex is 40%. The population that we were studying is sexually active adults aged 18-44 which we estimated to be roughly 75 million based on our research.

$$d = \frac{mr_{condom}}{Y}$$

where r_{condom} = the transmission probability of syphilis during protected sex. Based on our research, we estimated this to be around 40%.

$$g = \frac{1}{q}$$

where q = the amount of time between infection and diagnosis for a person with syphilis. This is the time when the person does not know that they are infected, and will continue to engage in sexual activity. We estimated this time to be 5 weeks or 35 days.

$$h = \frac{1}{w}$$

where w = the recovery time of a syphilis patient. This is the time where the recovering person would change their behavior and not have sex. After they recover, the person becomes susceptible again as they resume having sex. We estimated this to be about 3 weeks or 21 days based on our research.

b = # of new PrEP users per day (we assumed that this was constant value of 100 based on our research)

u = odds a PrEP user uses a condom

Model

$$\frac{dSPN}{dt} = -a * SPN * I + (1 - u) * b$$

$$\frac{dSNN}{dt} = -a * SNN * I - \frac{SNN}{SNN + SNP} * b$$

$$\frac{dSPP}{dt} = -d * SPP * I + \frac{SPP + SPN}{S} * h * R + u * b$$

$$\frac{dSNP}{dt} = -a * SNP * I + \frac{SNN + SNP}{S} * h * R - \frac{SNP}{SNN + SNP} * b$$

$$\frac{dI}{dt} = a * SPN * I + a * SNN * I + d * SPP * I + d * SNP * I - g * I$$

$$\frac{dR}{dt} = g * I - h * R$$

Initial Conditions

SPN0 = 216000

SNN0 = 45585511

SPP0 = 144000

SNP0 = 53513426

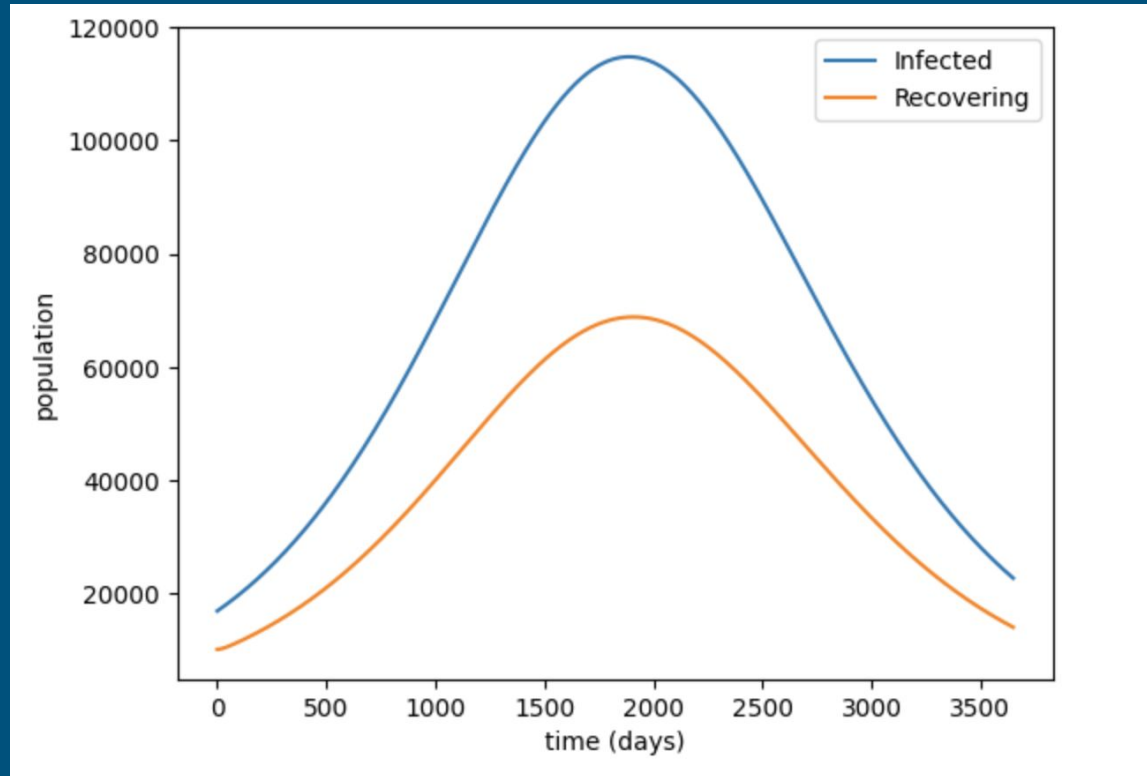
I0 = 16945

R0 = 10167

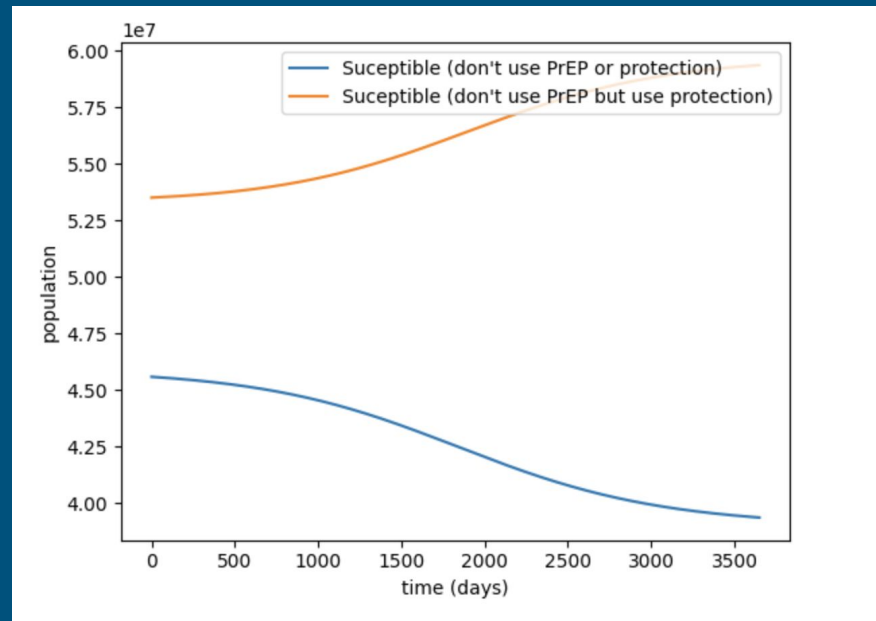
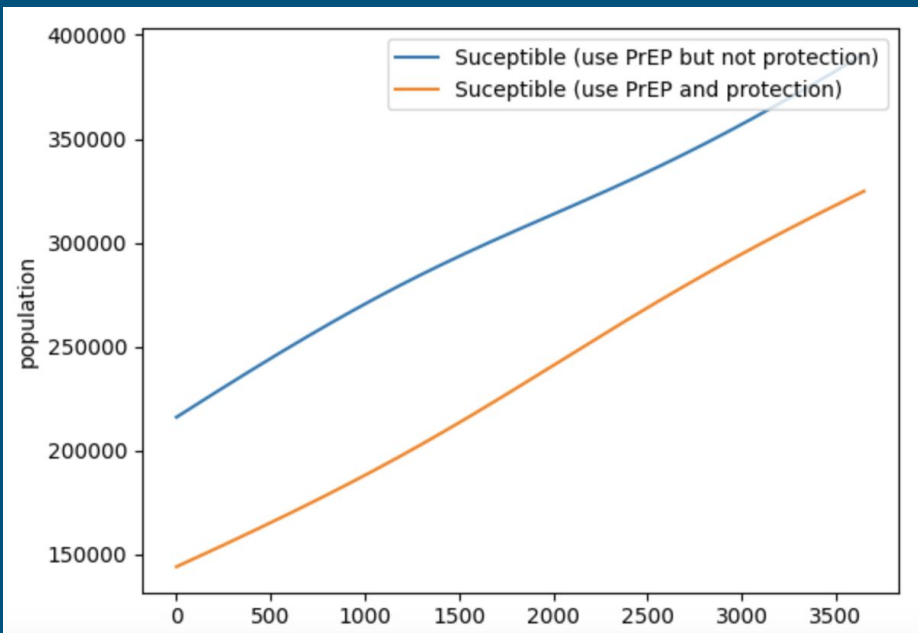
Condom usage rate:

- PrEP users: 40%
- Non-PrEP users: 54%

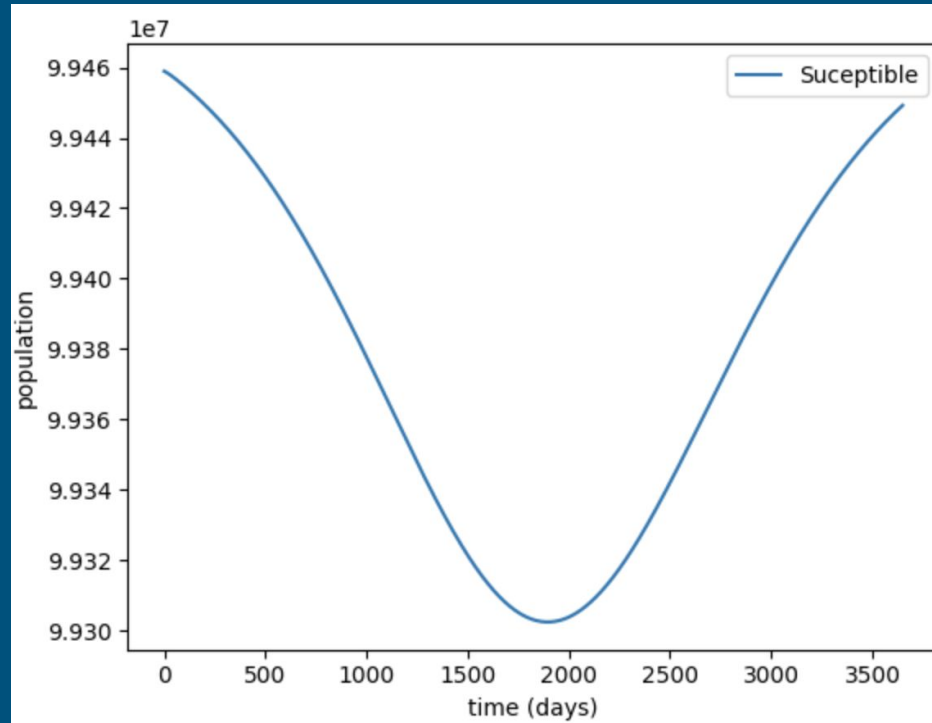
Results



Results



Results



Analysis

Equation modeling infected population has unstable fixed point at $I = 0$

- For populations of infected people around zero, the population of infected people will begin to grow

Whether or not people use protection has an impact on the rate of transmission which is seen in the population trends of the 4 susceptible groups

Conclusions

PrEP users not using protection does not seem to be the main contributor to the rise in syphilis cases

- Small percentage of the overall population
- Less likely to use condoms, but there is still a sizable chunk of the population that doesn't use PrEP that don't use protection
- Maybe should be studied in a more comprehensive model to predict the future if a larger percentage of the population goes on PrEP

Sources

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